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A Project Synopsis Report on

# Sensor Driven Adaptive Sanitation Protocol (SDASP)


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# ABSTRACT

Public sanitation remains a major concern in many parts of the world, particularly in high-traffic areas such as bus stations, railway stations, malls, and educational institutions. Despite the availability of public toilets, maintaining consistent hygiene standards is a persistent challenge. The primary issue lies in the reliance on manual cleaning schedules and periodic inspections, which fail to reflect the actual usage and real-time condition of the facilities. As a result, washrooms often become unhygienic between cleaning cycles, leading to foul odours, wet and slippery floors, and the accumulation of harmful gases such as ammonia ($NH_3$) and hydrogen sulphide ($H_2S$). These conditions not only create discomfort for users but also pose serious health risks, contributing to the spread of infectious diseases and respiratory issues. Additionally, the lack of monitoring systems results in poor accountability of maintenance staff and inefficient allocation of cleaning resources. To overcome these limitations, this project proposes a smart and automated solution known as the **Sensor Driven Adaptive Sanitation Protocol (SDASP)**. The system is designed to continuously monitor the hygiene conditions of public washrooms using a network of sensors and embedded systems. It integrates gas sensors to detect harmful gases, a moisture sensor to identify wet floor conditions, an infrared (IR) sensor to detect occupancy. These sensors collectively provide a comprehensive understanding of the washroom environment in real time. At the core of the system is the ESP32 microcontroller, which processes sensor data and makes intelligent decisions based on predefined threshold values. When the system detects unsafe or unhygienic conditions, it automatically activates response mechanisms such as exhaust fans through a relay module to improve air quality, and a buzzer to alert nearby staff. An LCD display is used to provide real-time status updates and alerts within the facility. Additionally, a keypad interface allows cleaning personnel to acknowledge alerts and log cleaning activities, thereby improving accountability and tracking. A key feature of the proposed system is its integration with IoT platforms, enabling remote monitoring and data visualization. The collected data is transmitted to the cloud, where it can be analysed to identify usage patterns, predict cleaning requirements, and optimize maintenance schedules. This shifts the sanitation management approach from a reactive system to a proactive and predictive model. Administrators can monitor multiple facilities remotely, receive instant notifications, and make informed decisions to ensure cleanliness.

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## Chapter-01

# INTRODUCTION

Maintaining hygiene in public washrooms is a persistent challenge, especially in densely populated regions where high usage leads to rapid deterioration of sanitation conditions. Despite increasing awareness about cleanliness and public health, a significant portion of public toilets still lack proper monitoring and maintenance systems. According to global reports, millions of people do not have access to clean and hygienic sanitation facilities, and a large percentage of existing public washrooms are rated in poor condition. This not only creates discomfort but also contributes to the spread of diseases such as cholera, typhoid, hepatitis, and respiratory infections. One of the major issues with current sanitation systems is their dependence on manual inspection and fixed cleaning schedules. Cleaning staff are typically assigned routine intervals, regardless of the actual condition of the washroom. This leads to inefficient use of resources, where some toilets are cleaned unnecessarily while others remain unhygienic for extended periods. Additionally, there is no reliable method to track staff accountability or verify whether cleaning tasks have been completed on time. The absence of real-time monitoring makes it difficult to ensure consistent hygiene standards.

To address these challenges, the proposed system, **Sensor Driven Adaptive Sanitation Protocol (SDASP)**, introduces an IoT-based smart solution for public washroom hygiene monitoring and management. The system leverages embedded technology and sensors to continuously monitor key environmental parameters such as gas concentration, floor wetness, and occupancy levels. Harmful gases like hydrogen sulphide ($H_2S$) and ammonia ($NH_3$), which are indicators of poor sanitation, are detected using advanced gas sensors. Simultaneously, moisture sensors identify wet or slippery floors, while infrared sensors track user movement to estimate occupancy.

The core of the system is powered by the ESP32 microcontroller, which processes sensor data and enables wireless communication. When any parameter exceeds predefined safety thresholds, the system automatically triggers appropriate actions such as activating exhaust fans, sounding alarms through buzzers, and displaying alerts on an LCD screen. Furthermore, the integration of a keypad allows cleaning staff to acknowledge alerts and log their presence, thereby ensuring accountability.

An important feature of SDASP is its cloud connectivity. Real-time data is transmitted to IoT platforms, enabling remote monitoring through dashboards. Administrators can track multiple washrooms, analyse trends, and receive instant notifications via messaging platforms. This transforms the system from a reactive model to a proactive and intelligent sanitation management solution.

## 1.1 OBJECTIVES

**1. Real-Time Environmental Monitoring**

To continuously monitor hydrogen sulfide ($H_2S$) and ammonia ($NH_3$) gas concentrations, floor moisture levels, and washroom occupancy using a network of embedded sensors connected to the ESP32-S microcontroller.

**2. Automated Threshold-Based Response**

To automatically trigger corrective actions — exhaust fan activation, buzzer alert, and LCD notification — whenever sensor readings exceed predefined safety thresholds, without requiring human detection of the problem.

**3. Staff Accountability and Alert Logging**

To ensure cleaning staff acknowledge every alert through a PIN-based keypad interface, creating a verifiable, time-stamped digital record of cleaning activity that eliminates unverified manual inspection.

**4. Remote IoT Monitoring and Notifications**

To transmit all sensor data and alert events to Web servers and notifications to supervisors, enabling multi-facility oversight without physical presence.

**5. Predictive and Data-Driven Sanitation Management**

To shift washroom maintenance from a reactive, schedule-based model to a proactive, data-driven approach by collecting historical sensor trends that enable pattern analysis and optimised cleaning schedules.

**6. Scalable Deployment Across Public Facilities**

To design the system as a scalable, low-cost solution suitable for deployment across diverse public environments including railway stations, schools, hospitals, malls, and Smart City infrastructure under the Swachh Bharat Mission.

## Chapter-02

# LITERATURE REVIEW

**Vimalesh Dhamale, Shekhar Singh, Shubham Zadane, and Manish Bhelande**, *"Smart Toilet Monitoring System Using IoT,"* International Journal of Computer Trends and Technology, vol. 68, no. 3, pp. 67–69, March 2020.

This paper [1] presents an IoT-based system designed to monitor the hygiene of public toilets using various sensors such as ammonia and gas sensors. The system collects data through a microcontroller and transmits it to a backend server for analysis and visualization. It also tracks cleaning activities to ensure proper maintenance. However, the system mainly focuses on monitoring and lacks advanced automation and intelligent decision-making features.

**Nikhil P. Mahalsekar, Shreyas Ganapaiah, Rakshith R. Poojary, Sushmitha, and Sathisha**, *"Intelligent Hygiene Monitoring System for Public Toilets,"* International Journal of Engineering Research & Technology (IJERT), vol. 11, no. 6, June 2022.

This paper [2] proposes an IoT-based centralized hygiene monitoring system that collects data related to air quality, water availability, and toilet usage. It provides real-time monitoring through a web interface and enables communication between administrators and cleaning staff. The system improves sanitation management and accountability but still depends on manual intervention for cleaning actions.

**K. Sai Kowshik, P. Laxmikanth, Mohammed Hassan, P. Sai Siddartha, and Ambika Naik Y**, *"A Review on Automated Hygiene for Public Toilet System Using IoT,"* TIJER – International Research Journal, vol. 10, issue 5, May 2023.

This review paper [3] discusses various IoT-based smart toilet systems that use sensors such as MQ-135 gas sensors, PIR sensors, ultrasonic sensors, and turbidity sensors. The study emphasizes real-time monitoring, improved hygiene, and efficient resource management. However, it mainly focuses on reviewing existing systems and lacks implementation of predictive or adaptive cleaning mechanisms.

**Nikita Ingale, Rutuja Sapate, Sayali Khose, and Veena Chitrashekhar**, *"Smart Public Restroom/Toilets Monitoring System,"* International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), vol. 4, issue 4, April 2024.

This paper [4] proposes an IoT-enabled air quality monitoring system for public restrooms using gas sensors, temperature sensors, and humidity sensors. The system includes an exhaust fan control mechanism and alert system to maintain proper air quality. It also supports real-time monitoring using cloud platforms. However, it mainly focuses on environmental monitoring and lacks features such as staff accountability and predictive analysis.

**F. I. Azman, N. L. Saleh, F. Hashim, A. Sali, A. M. Ali, and A. S. M. Noor**, *"An IoT-Based Hygiene Monitoring System in the Restroom,"* IEEE Access, vol. 13, 2025.

This paper [5] presents an advanced IoT-based hygiene monitoring system that integrates ammonia detection, humidity monitoring, and occupancy tracking. The system uses real-time data and threshold-based alerts to optimize cleaning schedules and improve efficiency. It also supports cloud-based data storage and long-term analysis using MQTT and time-series databases. However, the system involves higher complexity and cost compared to basic monitoring systems.

## 2.1 Summary on Literature Review

| Project Title | Journal / Conference | Authors | Year | Proposed System | Limitation |
|---|---|---|---|---|---|
| Smart Toilet Monitoring System Using IoT | IJCTT | Vimalesh Dhamale et al. | 2020 | IoT-based system using sensors and microcontroller to monitor gas levels, usage, and send data to a backend for analysis | Focuses mainly on monitoring; lacks automation and intelligent decision-making |
| Intelligent Hygiene Monitoring System for Public Toilets | IJERT | Nikhil P. Mahalsekar et al. | 2022 | IoT-based centralized system for monitoring air quality, occupancy, and water parameters with alert mechanisms | Limited automation; requires manual intervention for cleaning |
| A Review on Automated Hygiene for Public Toilet System Using IoT | TIJER | K. Sai Kowshik et al. | 2023 | Multi-sensor IoT system using gas, IR, ultrasonic, and turbidity sensors for hygiene monitoring | No predictive or adaptive cleaning; mostly theoretical/review-based |
| Smart Public Restroom/Toilets Monitoring System | IJARSCT | Nikita Ingale et al. | 2024 | IoT-based air quality monitoring system with gas sensors, temperature/humidity sensors, and ventilation control | Limited focus on staff accountability and advanced analytics |
| An IoT-Based Hygiene Monitoring System in the Restroom | IEEE Access | F. I. Azman et al. | 2025 | Advanced IoT system integrating ammonia detection, humidity, occupancy tracking, and data-driven cleaning alerts | Higher complexity and implementation cost |

## Chapter-03

# PROBLEM STATEMENT

The following core challenges motivate the design and development of SDASP:

- **Absence of Real-Time Monitoring:** Most public washrooms rely entirely on periodic manual inspection. There is no automated mechanism to detect elevated levels of harmful gases such as hydrogen sulfide ($H_2S$) or ammonia ($NH_3$), floor wetness from water spills, or current occupancy count.
- **Fixed Cleaning Schedules:** Cleaning is performed at predetermined time intervals regardless of the actual hygienic state of the facility. This results in under-cleaning when usage is high and over-cleaning when usage is low — both wasteful and ineffective.
- **No Staff Accountability:** Without a digital verification mechanism, there is no reliable way to confirm that cleaning staff have attended to an alert, completed a cleaning task, or been present in the washroom at the required time.
- **Disease Vector Risk:** Unhygienic washrooms are documented vectors for communicable diseases including malaria, typhoid, cholera, hepatitis A, and various respiratory illnesses caused by ammonia vapour and hydrogen sulfide exposure.
- **Manual Health Risk for Staff:** Manual inspection requires cleaning personnel to enter potentially hazardous environments without prior knowledge of gas concentration levels, posing direct health risks.
- **Lack of Data and Analytics:** In the absence of cloud-connected monitoring, no historical data is available for trend analysis, resource planning, or performance evaluation of cleaning staff.
- **High Maintenance Cost:** Traditional cleaning methods rely on fixed schedules and manual inspection, leading to inefficient use of resources such as water, electricity, and manpower. This increases overall operational costs without ensuring effective hygiene.
- **Delayed Response to Unhygienic Conditions:** In the absence of real-time monitoring, issues such as bad odour, gas accumulation, or wet floors are addressed only after complaints or scheduled inspections, resulting in delayed corrective actions.
- **User Dissatisfaction and Poor Experience:** Unclean and poorly maintained washrooms create a negative experience for users, reducing trust in public facilities and affecting the reputation of institutions managing them.

## Chapter-04

# BLOCK DIAGRAM

The SDASP system follows a clear Inputs → Processing → Outputs architecture with the ESP32-S microcontroller at its core. The block diagram below illustrates the complete signal flow and component interconnections within the system.

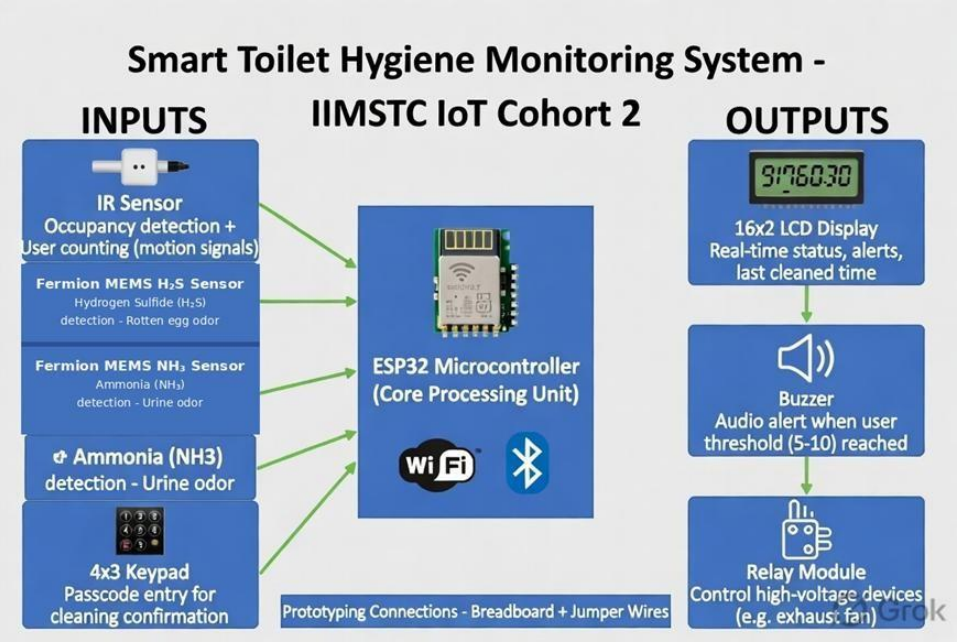



**Fig: Block Diagram**

## 4.1 Block Diagram Explanation

The system is divided into three functional layers:

### 1. INPUT LAYER:

- IR Sensor (FC-51): Positioned at the washroom entrance. Generates a digital HIGH/LOW signal each time a person crosses the infrared beam, incrementing or decrementing the occupancy counter.
- Fermion MEMS $H_2S$ Sensor: Continuously samples air. Outputs a voltage proportional to $H_2S$ concentration. ADC pin on ESP32 Dev kit V1 converts this to ppm values.
- Fermion MEMS $NH_3$ Sensor: Same principle as $H_2S$ sensor but calibrated for ammonia detection in the 5–500 ppm range relevant to urine odor.
- Humidity Sensor (DHT22): measures the relative humidity and temperature of the surrounding air. It is used to indirectly indicate wet or damp conditions by detecting an increase in humidity levels.
- 4×3 Membrane Keypad: 12-key input device for staff PIN entry. Scanned via 7 GPIO pins using row-column matrix scanning firmware.

**2. PROCESSING LAYER — ESP32 Dev kit V1:**

- All input signals converge at the ESP32 Dev kit V1. The firmware runs a continuous polling loop: read sensors → compare against thresholds → trigger outputs if needed → transmit data to cloud → repeat.
- The ESP32 Dev kit V1's dual-core architecture allows simultaneous execution of sensor polling (Core 0) and Wi-Fi transmission (Core 1) without blocking.

**3. OUTPUT LAYER:**

- 16×2 LCD Display: Shows real-time system status, sensor values, alert messages, and prompts the staff to enter PIN.
- Relay Module: Driven by a GPIO pin on ESP32 Dev kit V1. When energized, it closes the circuit for the exhaust fan (AC mains), running ventilation automatically.
- Active Buzzer: Emits audible alert tone on threshold breach. Deactivated only after correct PIN entry by staff.
- Cloud: Receives MQTT/HTTP POST packets from ESP32 Dev kit V1 over Wi-Fi. Stores time-series data and hosts the admin dashboard.

## Chapter-05

# METHODOLOGY

SDASP implements a 7-phase operational protocol that executes continuously in the ESP32 Dev kit V1main loop. Each phase builds upon the previous, creating a complete sensor-to-action pipeline:

**1. System Initialization**
On power-up, the ESP32 Dev kit V1 initializes all GPIO pins for sensors, relay, buzzer, and keypad. The $I^2C$ bus is started and the LCD is configured to display 'System Ready'. Fermion MEMS gas sensors require a ~20-second preheating period for electrochemical cell activation and calibration. During this window, the firmware enters a countdown loop with status displayed on the LCD. Wi-Fi connection to the configured is established and cloud platform credentials are verified.

**2. LCD Status Display**
Throughout all subsequent phases, the LCD continuously updates to reflect the current system state. Line 1 displays real-time sensor values ($H_2S$ ppm and $NH_3$ ppm alternating every 2 seconds). Line 2 shows the current alert status, occupancy count, or PIN entry prompt depending on the active phase.

**3. Occupancy Monitoring (IR Sensor)**
The IR sensor is mounted at the washroom entrance at a height of approximately 80 cm (hip level) to detect pedestrian crossings reliably. The firmware implements a simple counter with debounce logic: each entry event increments the occupancy integer; each exit event decrements it (minimum 0). This prevents false gas alerts when the washroom is unoccupied — an important design decision to avoid nuisance alarms from residual gas dissipation after use.

**4. Gas Level & Moisture Sampling**
The ESP32 ADC reads the analogue output voltages from the $H_2S$ sensor (GPIO34) and $NH_3$ sensor (GPIO35) every 500 ms. Raw ADC counts are converted to ppm using a calibration formula derived from the sensor datasheets. Simultaneously, the YL-69 moisture sensor output on GPIO32 is read. A reading below a configurable threshold voltage triggers a 'Floor Wet' condition flag.

**5. Automated Response (Relay + Buzzer)**
If $H_2S > 5$ ppm OR $NH_3 > 5$ ppm OR floor moisture flag is set, the system enters ALERT state: (a) GPIO5 drives the relay module LOW, closing the exhaust fan circuit and activating ventilation. (b) GPIO18 drives the buzzer HIGH through the transistor driver, emitting a continuous 2.5 kHz tone. (c) LCD displays the specific alert condition and the message 'ENTER PIN'. The relay remains ON until gas levels normalize and a correct staff PIN has been entered.

**6. Staff Acknowledgement (4×3 Keypad)**
Upon alert activation, the firmware initiates a keypad scanning loop. The membrane keypad is scanned using a row-column multiplexing technique: 4 row pins are driven LOW sequentially while 3 column pins are read with internal pull-ups enabled. The entered PIN is compared against a stored reference PIN. An incorrect PIN resets the buffer and re-prompts. A correct PIN deactivates the buzzer, logs the acknowledgement event with timestamp to cloud, and resets the alert state (relay remains ON until gas normalizes).

## 7. Cloud Logging & Remote Alerts

After each complete sensor cycle, the ESP32 Dev kit V1 transmits a JSON payload over Wi-Fi via HTTP POST to API channels: $H_2S$ ppm, $NH_3$ ppm, moisture level, occupancy count, alert status, and timestamp. The Blynk IoT platform provides a mobile dashboard with real-time gauge widgets. A Telegram Bot API call is triggered on every alert event, pushing an instant message with alert details to a configured group chat accessible by supervisors. Future integration with WhatsApp Business API (via Twilio) is planned.

# Chapter-06

# HARDWARE AND SOFTWARE REQUIREMENTS

## 6.1 Hardware Components

| # | Component | Model / Part No. | Function | Key Specification |
|---|---|---|---|---|
| 1 | ESP32 DevKit V1 | ESP32-S (30-pin) | Central controller, Wi-Fi, ADC, GPIO | Dual-core 240 MHz, 520 KB SRAM, 802.11 b/g/n Wi-Fi, 18× ADC channels |
| 2 | $H_2S$ Gas Sensor | Fermion MEMS $H_2S$ Sensor | Detects hydrogen sulfide (rotten-egg odor) | Range: 1–200 ppm; Response time: <30 s; Analogue output |
| 3 | $NH_3$ Gas Sensor | Fermion MEMS $NH_3$ Sensor | Detects ammonia / urine odour | Range: 5–500 ppm; MEMS electrochemical cell; Analogue output |
| 4 | IR Sensor | FC-51 / TCRT5000 | Entry/exit occupancy counting at door | Operating distance: 2–30 cm; Digital GPIO output; 3.3/5 V compatible |
| 5 | Humidity Sensor | DHT22 | Floor wetness and spill detection | Analogue + digital output. |
| 6 | 16×2 LCD Display | JHD162A + $I^2C$ Backpack | Status display and alert messages | 16 columns × 2 rows; $I^2C$ interface (SDA/SCL); 5 V; Contrast adjustable |
| 7 | 4×3 Matrix Keypad | Generic Membrane 4×3 | Staff PIN entry to acknowledge alerts | 12 keys (0–9, *, #); 7-pin GPIO; Membrane type; <₹30 cost |
| 8 | Relay Module | SRD-05VDC-SL-C | Switches exhaust fan ON/OFF on threshold breach | 5 V coil; 250 VAC / 10 A capacity; Opto-isolated input |
| 9 | Active Buzzer | Active Piezo 5V | Audible alert when threshold is exceeded | 5 V active buzzer; ~85 dB SPL; 2.5 kHz tone; No external oscillator |
| 10 | Breadboard + Jumper Wires | Generic | Prototyping and interconnection | 830-point breadboard; Male-to-male and male-to-female jumper wires |

### 6.1.1 Components Overview

1. **ESP32 DevKit V1 (Microcontroller)**

The **ESP32 DevKit V1** is the main microcontroller used in the system, acting as the central processing unit. It is based on the ESP32 chip, which features a dual-core processor, built-in Wi-Fi and Bluetooth connectivity, and multiple GPIO pins for interfacing with sensors and output devices. In this project, the ESP32 Dev kit V1 collects data from sensors such as gas sensors, humidity sensor, and IR sensor, processes the data in real-time, and controls components like the relay, buzzer, and LCD display based on predefined conditions. Additionally, its Wi-Fi capability enables transmission of data to cloud platforms for remote monitoring and alert systems, making the solution IoT-enabled and efficient for smart sanitation management.

2. **$H_2S$ Gas Sensor (Fermion MEMS)**

The **$H_2S$ gas sensor** is used to detect hydrogen sulfide gas, which is responsible for the characteristic “rotten egg” smell in unclean washrooms. It continuously monitors gas concentration in parts per million (ppm). When the gas level exceeds the predefined threshold, the system triggers alerts and activates ventilation. This helps maintain air quality and hygiene.

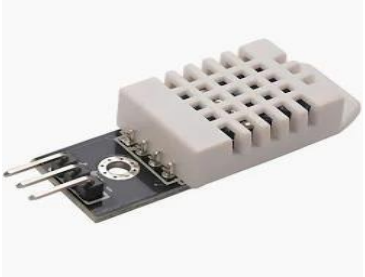
3. **$NH_3$ Gas Sensor (Fermion MEMS)**

The **$NH_3$ sensor** detects ammonia gas, commonly found due to urine odor in washrooms. It provides real-time monitoring of ammonia levels and helps identify poor sanitation conditions. If the concentration crosses the safe limit, the system generates alerts and initiates corrective actions such as turning on exhaust fans.

**4. IR Sensor (FC-51 / TCRT5000)**

The **Infrared (IR) sensor** is used for occupancy detection. It is installed near the entrance to count the number of people entering and exiting the washroom. This helps in maintaining an occupancy count and avoids false alerts when the washroom is not in use. It operates using digital signals and detects objects within a short range.

**5. Humidity Sensor (DHT22)**



The DHT22 humidity sensor is used to measure the relative humidity and temperature of the surrounding environment. In this system, it is utilized to indirectly indicate wet floor conditions, as the presence of water on the floor increases the moisture content in the air, leading to higher humidity levels. When the detected humidity exceeds a predefined threshold, the system interprets it as a possible sign of damp or wet conditions and triggers an alert. However, this approach provides only an indirect indication and may be influenced by external factors such as ventilation and temperature variations.

**6. 16×2 LCD Display (JHD162A with I²C)**

The **16×2 LCD display** is used to show system status and alert messages. It displays information such as “System Ready,” gas levels, occupancy count, and warning alerts. The I²C interface reduces wiring complexity by using only two communication lines (SDA and SCL), making it efficient for embedded systems.

### 7. 4×3 Matrix Keypad

The **4×3 keypad** is used for staff authentication. It allows maintenance staff to enter a secure PIN to acknowledge alerts and confirm cleaning actions. This ensures accountability and prevents unauthorized dismissal of alerts. It is cost-effective and easy to interface with the ESP32 using GPIO pins.

### 8. Relay Module (SRD-05VDC-SL-C)

The **relay module** acts as an electrically operated switch. It is used to control high-power devices like exhaust fans. When gas levels exceed the threshold, the ESP32 activates the relay to turn ON the fan, improving air circulation. The relay provides isolation between low-power control circuits and high-power devices.

### 9. Active Buzzer (5V)

The **active buzzer** provides an audible alert when unsafe conditions are detected. It produces a sound automatically when powered, without requiring an external oscillator. This ensures immediate attention from maintenance staff when gas levels or moisture levels exceed safe limits.

**10. Breadboard and Jumper Wires**

The **breadboard and jumper wires** are used for prototyping the circuit without soldering. They allow easy connection and testing of components during development. This setup helps in modifying the circuit design quickly before final PCB implementation.

## 6.2 Software Tools

### 6.2.1 Arduino IDE

The **Arduino IDE (Integrated Development Environment)** is an open-source software platform used for writing, compiling, and uploading code to microcontrollers such as the ESP32. It supports programming in C/C++ and provides a simple interface with features like code editing, serial monitoring, and library management. In this project, the Arduino IDE is used to interface sensors, process data, and control output devices like the relay, buzzer, and LCD, enabling smooth implementation of the system.

### 6.2.2 Web Dashboard

The software module of the Sensor-Driven Adaptive Sanitation Protocol (SDASP) project is developed as a Web-Based Monitoring Dashboard to provide real-time monitoring, user authentication, staff accountability, and system status visualization. The dashboard helps users and maintenance staff easily understand the hygiene condition of the toilet without checking raw sensor values directly. It displays important parameters such as occupancy status, user count, air quality status, floor wetness, humidity level, exhaust fan condition, and system alerts in a simple and user-friendly format.

The dashboard includes a login system with role-based access, where both Admin and Staff users can log in. Admin has full monitoring access, while staff can update cleaning activities through the accountability section. A staff activity tracking feature records who performed cleaning and when it was done, improving maintenance responsibility and monitoring.

The system also uses smart alert logic, where occupancy duration and inactivity are monitored to detect abnormal situations such as long occupancy or no movement, indicating possible emergency conditions. Based on threshold conditions, the dashboard displays NORMAL, WARNING, or EMERGENCY status using color indicators for easy understanding.

The web dashboard is designed to be lightweight so that it can be hosted directly on the ESP32 Dev kit V1 using SPIFFS or LittleFS, making it suitable for real-time IoT implementation.

### 6.2.3 Language Used

- **HTML (HyperText Markup Language)** HTML is used to create the basic structure of the web dashboard. It defines the login page, monitoring cards, accountability section, buttons, status displays, and overall layout of the system interface.
- **CSS (Cascading Style Sheets**) CSS is used for designing the visual appearance of the dashboard. It provides professional UI styling, responsive card layout, color indicators (green/yellow/red), login page design, spacing, alignment, and overall dashboard attractiveness.
- **JavaScript** JavaScript is used to implement the functional logic of the dashboard. It handles login authentication, role-based access control, dynamic sensor value updates, emergency detection logic, occupancy tracking, activity logging, and real-time status changes.

## Chapter-07

# EXPECTED OUTPUT

### 1. Functional Smart Washroom Prototype

A working prototype integrating the ESP32 Dev kit V1 microcontroller with all sensors ($H_2S$, $NH_3$, IR, moisture), output devices (relay-controlled exhaust fan, buzzer, LCD), and a PIN-based keypad for staff acknowledgement. The system automatically detects unsafe conditions, triggers alerts, logs staff responses, and transmits all data to cloud dashboards in real time.

This prototype validates the end-to-end SDASP pipeline — from sensor detection through automated response to cloud logging — demonstrating a shift from manual, schedule-based washroom maintenance to an intelligent, data-driven sanitation management system.

### 2. ESP32-Hosted Web Server for Real-Time

A lightweight web server built and hosted directly on the ESP32 Dev kit V1, accessible to any device on the same network via a local IP address. The web interface displays live sensor readings, current alert status, occupancy count, and the staff cleaning log — eliminating the need for third-party cloud platforms or external dashboard subscriptions.

### 3. Automated Alert and Actuator Control System

When sensor readings breach predefined thresholds, the ESP32 Dev kit V1 simultaneously triggers the exhaust fan relay, activates the buzzer, and updates the alert status on the web server interface
— all without manual intervention. The actuator states are also reflected live on the hosted webpage, so remote supervisors can confirm that corrective action has been taken.

The system holds the alert state active until gas levels return to safe levels and a staff PIN acknowledgement is received, ensuring that alerts are not dismissed without physical attendance. This closed-loop control mechanism guarantees accountability at both the hardware and software level.

### 4. PIN-Based Staff Acknowledgement and Cleaning Log

Cleaning staff acknowledge every alert by entering a PIN through the keypad, which the ESP32 Dev kit V1 validates and records with a timestamp. The acknowledgement entry is immediately reflected on the web server, updating the cleaning log visible to supervisors in real time without any additional cloud dependency.

This log serves as a digital replacement for paper-based inspection registers, providing facility managers with a verifiable, time-stamped record of every alert event and staff response. Over time the log builds a reliable dataset of cleaning patterns, response times, and recurring problem periods across the facility.

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